

Data Structures and Algorithms – Concepts & Subtopics (Java Oriented)

1. Foundations of DSA

Time & Space Complexity

- Algorithm efficiency
- Input size (n)
- Time complexity
- Space complexity
- Best, Worst, Average case
- Big-O, Big-Ω, Big-Θ
- Constant, Linear, Logarithmic, Polynomial, Exponential
- Amortized analysis
- Time vs Space trade-off

Recursion

- Recursion definition
- Base case & recursive case
- Call stack
- Stack frame
- Recursion depth
- Stack overflow
- Tail recursion (concept)
- Recursion vs Iteration
- Advantages & disadvantages

2. Linear Data Structures

Array

- Static vs Dynamic arrays
- Memory allocation
- Contiguous memory
- Indexing & random access
- Traversal
- Insertion, Deletion, Update (concept)
- Time complexity
- Advantages & limitations
- Use cases

String (Java)

- String definition
- Character array vs String
- Immutability
- String Constant Pool
- Literal vs new keyword
- equals() vs ==
- StringBuilder
- StringBuffer
- Performance impact

Linked List

- Node structure
- Singly, Doubly, Circular linked list
- Head & Tail
- Traversal
- Insertion & Deletion (concept)
- Searching
- Time complexity
- Advantages & disadvantages
- Use cases

Stack

- LIFO principle
- Push, Pop, Peek
- Overflow & Underflow
- Array vs LinkedList stack
- Call stack
- Applications
- Limitations

Queue

- FIFO principle
- Enqueue & Dequeue
- Front & Rear
- Linear queue
- Circular queue
- Deque
- Priority queue (concept)
- Applications
- Limitations

3. Non-Linear Data Structures

Tree Basics

- Node, Edge, Root
- Parent, Child, Sibling
- Leaf & Internal node
- Degree
- Level
- Height vs Depth
- Path
- Subtree

Binary Tree

- Binary tree definition
- Full, Complete, Perfect, Skewed trees
- Inorder, Preorder, Postorder traversal
- Level order traversal
- Applications

Binary Search Tree (BST)

- BST property
- Search, Insert, Delete (concept)
- Min & Max element
- Successor & Predecessor
- Balanced vs Unbalanced

- Time complexity
- Limitations

Heap

- Heap definition
- Complete binary tree property
- Heap order property
- Min Heap vs Max Heap
- Insertion & Deletion
- Heapify concept
- Applications
- Heap vs BST

4. Graphs

Graph Basics

- Vertex & Edge
- Directed vs Undirected
- Weighted vs Unweighted
- Cyclic vs Acyclic
- Connected vs Disconnected
- Self loop & Parallel edges

Graph Representation

- Adjacency Matrix
- Adjacency List
- Space & Time comparison
- Use cases

Graph Traversal

- Breadth First Search (BFS)
- Depth First Search (DFS)
- Queue usage in BFS
- Stack/Recursion in DFS
- Applications

5. Searching Algorithms

Linear Search

- Sequential access
- Time complexity
- Use cases
- Limitations

Binary Search

- Sorted data requirement
- Divide and conquer idea
- Mid element calculation
- Iterative vs Recursive
- Time complexity
- Limitations

6. Sorting Algorithms

Basic Sorting

- Bubble Sort
- Selection Sort
- Insertion Sort
- Working principle
- Time & Space complexity
- Use cases

Advanced Sorting

- Merge Sort
- Quick Sort
- Heap Sort
- Divide & Conquer
- Best, Average, Worst case
- Stability

Sorting Concepts

- Stable vs Unstable
- In-place vs Out-of-place
- Comparison based sorting
- Non-comparison sorting (concept)

7. Hashing

Hashing Fundamentals

- Hash function
- Hash table
- Key-value pair
- Load factor
- Rehashing

Collision Handling

- Separate chaining
- Linear probing
- Quadratic probing
- Double hashing
- Pros & Cons

Java Hashing Internals

- HashMap internal working
- hashCode() & equals()
- Bucket concept
- Treeification
- Time complexity
- HashMap vs Hashtable
- HashMap vs ConcurrentHashMap
- HashMap vs TreeMap

8. Algorithm Design Paradigms

- Brute force
- Divide & Conquer
- Greedy approach
- Backtracking (concept)

- Dynamic Programming (concept)
- Optimal substructure
- Overlapping subproblems

9. Bit Manipulation

- Binary number system
- AND, OR, XOR, NOT
- Left shift & Right shift
- Signed vs Unsigned shift (Java)
- Bit masking
- Applications

10. Java-Specific Concepts

Memory Management

- Stack memory
- Heap memory
- Object creation
- Reference variables
- Garbage collection overview
- Memory leaks (concept)

Collections Framework

- Collection hierarchy
- List, Set, Map interfaces
- ArrayList vs LinkedList
- HashSet vs TreeSet
- HashMap vs TreeMap
- Time complexity comparison
- Use cases